

Manepalli Mohan Kumar

Curriculum Vitae — AI Engineer, Medical Imaging

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A results-driven **AI Engineer** with **3+ years** of hands-on experience in **Deep Learning** and **Machine Learning** (2023 — present), including **1.5 years** of professional experience in AI for **Medical Imaging**. Currently working as **Technical Staff at TANUH (AICoE), IISc**, developing **vascular disease screening** models that detect **systemic diseases** from fundus images. I build, fine-tune, and debug **PyTorch** models for **medical image segmentation**, classification, **denoising**, and **image translation**, including foundation-model fine-tuning (**RETFound**, **MedGemma**) with **LoRA**. My interests include **systemic disease detection** and **fundus & OCT imaging**.

EDUCATION

Bachelor of Technology 2022 — APRIL 2026

Rajiv Gandhi University of Knowledge Technologies

CGPA: 8.5

Pre-University Course (Intermediate / MPC)

2020 — 2022

Rajiv Gandhi University of Knowledge Technologies

CGPA: 9.56

Secondary School Certificate (10th Class)

2020

BSE AP

- Secured the **State 64th Rank** in the 2020 Polytechnic Entrance Examination immediately following 10th grade.

PUBLICATIONS & PRESENTATIONS

"Energy-Regularized Self-Supervised Denoising for Structure-Preserving OCT Imaging"

MARCH 2026

Optics Letters Journal

Published

- Addressed the **over-smoothing problem** in **S2SNet** for the **OCT dataset** by introducing energy regularization and validated improvements through a comprehensive comparison of **segmentation performance**.

"Data Driven Approach in Building Consumer-Friendly Business Models Using Machine Learning Techniques"

DEC 2024

IEEE AICECS Conference

Manipal Institute of Technology

- Presented a research paper comparing three machine learning models on real-time dataset metrics.

PROFESSIONAL EXPERIENCE

Technical Staff / TANUH (AICoE) IISc

JUNE 2026 — PRESENT

Vascular Disease Screening Vertical

Bangalore, KA

- Working in the **Vascular disease screening** vertical to detect **systemic diseases** from only **4 fundus images** per patient, leveraging **RETFound**, **MedGemma 1.5**, and **Qwen 3.8 27B** with **LoRA** fine-tuning.
- Built an **agent-based automated data quality check pipeline** to validate incoming clinical imaging data at scale.
- Developed and operated a **data collection and anonymisation website** (developer & IT owner) hosted on **Google Cloud Platform (GCP)** across India for **real-time data collection from hospitals**.
- Administered the team's **compute nodes** and **Open OnDemand** software, managing access and workloads for researchers.

Engineering Intern / TANUH (AICoE) IISc

NOV 2025 — MAY 2026

Medical Assistant and Early Disease Detection

Bangalore, KA

- Working at **TANUH (AI Centre of Excellence under the Ministry of Education)** to develop **AI-driven diagnostics** and screening tools for early **breast cancer detection** and **risk prediction**.
- Developing **"Pseudohealth"** generative systems to synthesize **mammogram** images and **finetuned MedGemma 1.5 4B** to work with **mammogram** images, transforming clinical care through scalable AI solutions.

Developer / Video Masking & Annotation Application

AUG 2025 — NOV 2025

Full-Stack AI Deployment

Bangalore, KA

- Designed a medical video annotation tool integrating the **Segment Anything Model (SAM)** for frame-by-frame analysis, deployed via a scalable serverless architecture on **Google Cloud Run**.

Research Intern / MIGLab [Link], CDS Department at IISc

MARCH 2025 — OCT 2025

Medical Imaging & Deep learning

Bangalore, KA

- Executed distributed training via **torchrun** across **DGX Spark** clusters and optimized model inference through **Quantization** for real-time serving.
- Implemented **Flash Attention** and **Maximum Representative Subsampling (MRS)** pipelines for the **Federated Tumor Segmentation (FeTS) dataset** to train **3D UNet** models, optimizing data quality and training performance for large-scale vision tasks.
- Developed **OCT image denoising** algorithms and currently **developing model-based Quantitative Susceptibility Mapping (QSM) reconstruction** algorithms, focusing on **preserving structural details** and high-precision accuracy.

- Supported instruction and technical experiments for a course on **AI in Healthcare** focused on **medical image analysis**.

Machine Learning Intern / Academor (Flutura)

JUNE 2023 — AUG 2023
Remote

- Gained professional experience in applied machine learning, focusing on model internals, data curation, and version-controlled experimentation for industrial use cases.

ACADEMIC PROJECTS

Deployment / Real-time Median-nerve Segmentation

JUNE 2025 — AUG 2025

- Deployed **VisTr** on **Jetson Orin NX** using **NVIDIA SDK Manager**, while optimizing transformer performance with **Flash-Attn 2** and scaling training via **PyTorch DDP** on **dual RTX A6000 GPUs**.

Experimentation / Stable Diffusion 3.5 Medium & ComfyUI

2024

- Executed inferencing of **Stable Diffusion 3.5** to generate high-quality images from text prompts utilizing the **ComfyUI** node-based interface on **Kaggle** infrastructure.

Developer / AgriBot (Collab with ICAR, Pune)

JAN 2024 — AUG 2024

- Developed an end-to-end multimodal application for leaf disease detection and crop recommendation, integrating vision capabilities with **Arduino**, **ESP**, and **Raspberry Pi** for real-time inference.

Developer / Gemini AI-Powered Telegram Bot

DEC 2023

- Built a multimodal agentic chatbot integrated with Google’s Gemini API, utilizing **precise model configuration** for domain-specific user query answering.

Developer / Cloud-Based Automatic Object Finder

2023

- Engineered a deep learning system for real-time multimodal object identification, utilizing cloud-streamed data and **YOLO** architectures.

Developer / LI-FI Prototype

2023

- Developed a hardware-integrated prototype for high-speed wireless data transmission via light, demonstrating technical versatility in hardware-AI systems.

TECHNICAL SKILLS

Programming & Frameworks	Python (PyTorch, PyTorch Lightning, TensorFlow, Scikit-learn, MONAI), MATLAB, OpenCV, C/C++, HF Transformers, ComfyUI
LLMs & Generative AI	Vision Language Models (VLMs) & Large Language Models (LLMs) — basic knowledge with some hands-on experience; Diffusion Models (DDPM), VAEs, GANs, Fine-tuning
DL Architectures	Transformer-based models (ViTs), Flash Attention, UNet, Multi-task & Multi-modal Learning
Imaging Systems & Infra.	Medical Imaging (MRI, CT, Fundus, OCT), DICOM/NIfTI formats, Model Optimization, Edge/Mobile Architectures
CV & Processing	Image/Video Restoration, Enhancement, Denoising, Segmentation, Object Detection, OCR, Feature Extraction
Distributed Training & Infra.	DeepSpeed, torchrun, PyTorch, PyTorch Lightning DDP, GPU Acceleration, bfloat16, DGX Spark Clusters
Inference Optimization	Model Quantization (GGUF), Knowledge Distillation, vLLM, TensorRT
Algorithms & Concepts	Regression, Decision Trees, Random Forest, Gradient Boosting, Clustering, PCA, Neural Networks, CNNs
Experiment Tracking & DevTools	MLflow, TensorBoard, Python Logger, Git CLI
Communication	English, Telugu (Native), Hindi, Technical Documentation

CERTIFICATIONS

Machine Learning for Engineering: NPTEL / IIT Madras [Link]	2025	Circuits & Electronics 3: edX / MITx [Link]	2024
VLSI Design Flow: RTL to GDS: NPTEL / IIIT Delhi [Link]	2025	Geoprocessing using Python: IIRS / ISRO [Link]	2023
Electronic Systems Design (PCB): NPTEL / IIT Madras [Link]	2025	Intro to Machine Learning: Kaggle [Link]	2023
Python (Basic) Certified: HackerRank [Link]	2025	AI Appreciate & AI Aware (Combined): Intel [Link]	2023
Essential Mathematics for ML: NPTEL / IIT Roorkee [Link]	2024	Pre-University Course (Intermediate / MPC): RGUKT [Link]	2022
Deep Learning Specialist (PyTorch, TensorFlow, Keras): edX / IBM [Link]	2024	Secondary School Certificate (10th Class): BSE AP [Link]	2020

ACTIVITIES

- Regularly analyze **Q1/Q2 journals** and state-of-the-art (**SOTA**) research to stay at the forefront of AI advancements.
- Participant: IEEE YESIST’12 Competition [\[Link\]](#)
- National Service Scheme (NSS)

2024
2021 — 2024